$$= 1.8 \times 10^{9} \text{ J}$$

4(a) the *uniform magnetic flux density* which when acting *perpendicularly* to a straight conductor carrying

a current of *1 ampere* produces a force per unit length of *1 Newton per metre* on the conductor

4(b)(i)

$$\left.\begin{aligned} q &= 2e = 2\times\left(1.6\times10^{-19}\right) = 3.2\times10^{-19} \text{ C} \\ m &= 4u = 4\times\left(1.66\times10^{-27}\right) = 6.64\times10^{-27} \text{ kg} \end{aligned}\right\}$$

Qns

4(b)(ii)1. magnetic force provides centripetal force

$$Bqv_T = \frac{mv_T^2}{r}$$

$$v_T = \frac{Bqr}{m} = \frac{(0.519)(3.2\times10^{-19})(0.02)}{(6.64\times10^{-27})}$$

$$= 5.0\times10^5 \text{ m s}^{-1}$$

4(b)(ii)2. magnetic force provides centripetal force, $Bqv_T = mr\omega^2$

$$Bqv_T = mr\left(\frac{2\pi}{T}\right)^2$$

$$T = 2\pi\sqrt{\frac{mr}{Bqv_T}}$$

$$= 2\pi\sqrt{\frac{(6.64\times10^{-27})(0.02)}{(0.519)(3.2\times10^{-19})(5.0\times10^5)}}$$

$$= 2.51\times10^{-7} \text{ s}$$

4(b)(ii)3.

$$\tan 60^\circ = \frac{v_T}{v_Z}$$

$$v_Z = \frac{5.0\times10^5}{\tan 60^\circ}$$

$$= 2.89\times10^5 \text{ m s}^{-1}$$

4(b)(ii)4. $p = v_Z t$

$$= (2.89\times10^5)(2.51\times10^{-7})$$

$$= 0.0725 \text{ m}$$

4(b)(iii) helical path with <u>decreasing pitch</u>
alpha particle experience deceleration due to electric force acting on it in direction opposite to v_Z

5(a)(i) slit width = slit width